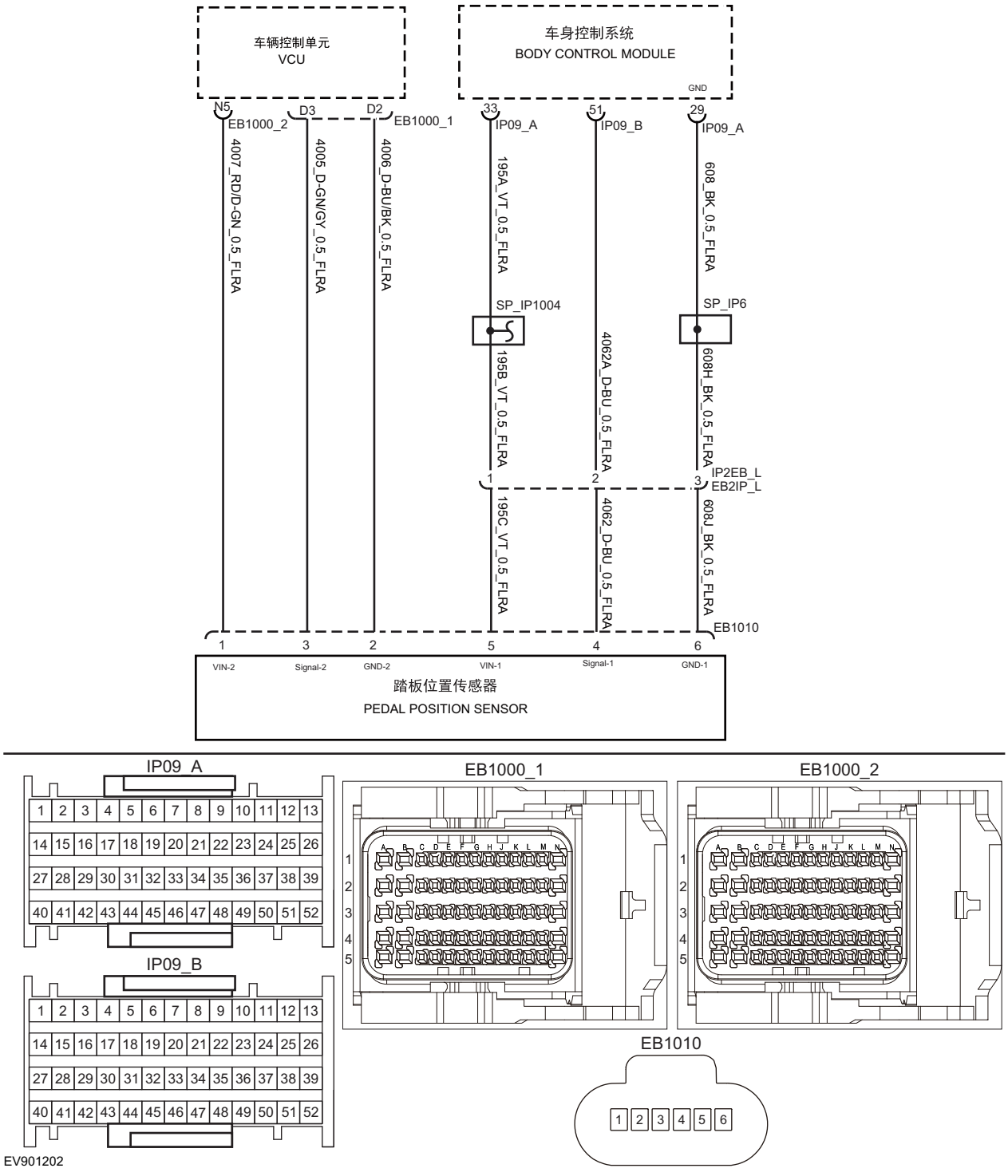
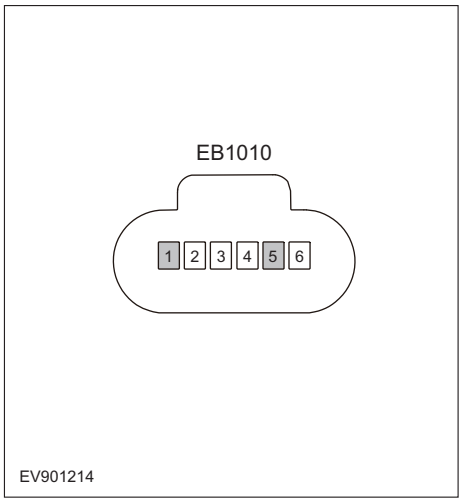
P057D、 P057C、 P057B、 P0571

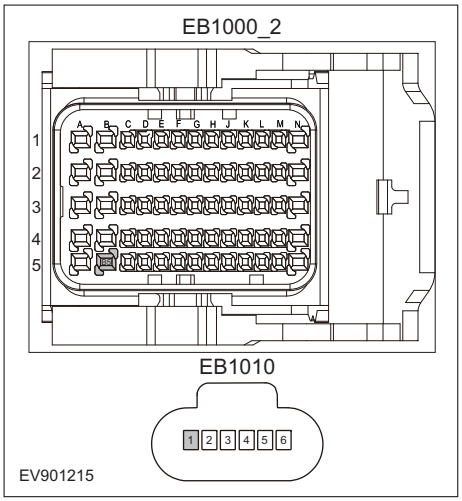
Fault diagnosis code
P057D：DEM_stBkPedlSenSBCfmd
P057C：DEM_stBkPedlSenSGCfmd
P057B：DEM_stBkPedlUSenOrng
P0571：DEM_stBkPedlPlaus
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • VCU module fault • Brake pedal positioning sensor fault.
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P057D
Brake pedal positioning sensor voltage is greater than or equal to 4.9V
P057C
Brake pedal positioning sensor voltage is lower than or equal to 4.9V
P057B
Any of the following conditions are satisfied: 1, Brake pedal positioning sensor voltage is greater than 4.5V ; 2, Brake pedal positioning sensor voltage is lower than 4.9V ;
P0571
The absolute value of the difference between the brake pedal opening and VCU’s brake pedal opening is greater than or equal to 10%
To set the effect of a fault code condition
Description of circuit diagram
The VCU module monitors the brake pedal position sensor via low level reference circuit and signal circuit.

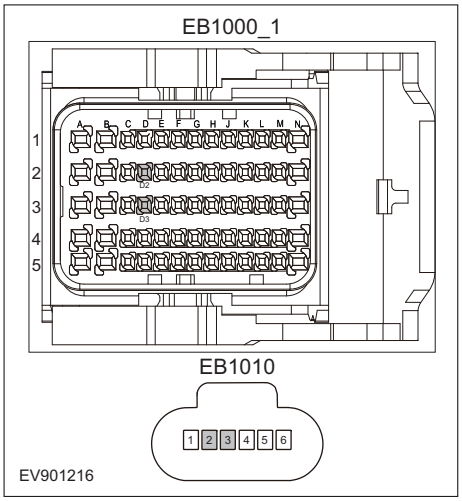
Circuit diagram



Troubleshooting steps	
Step 1	General inspection.
	1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of brake pedal position sensor,VCU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the VCU with diagnostic apparatus. 4 Is there any other fault code except for P057D、 P057C、P057B、 P0571?
	YES → Refer to:DTC Summary list(VCU).
NO ↓	
Step 3	Check the supply power of brake pedal positioning sensor when it is started or operating.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1010 of brake pedal positioning sensor. 3 Turn the ignition switch to ON. 4 Measure the voltage between the plug terminal of brake pedal positioning sensor and ground. Measuring circuit: voltage between terminal 1 on plug EB1010 and ground. Measuring circuit: voltage between terminal 5 on plug EB1010 and ground. **Standard value:4.7~5.2V** 5 Check whether the result is normal or not?
	NO → Repair the circuit.
YES ↓	

Step 4	Check the circuit between brake pedal positioning sensor and VCU (open).





1 Turn the ignition switch to OFF.

2 Disconnect the harness plug EB1000_1, EB1000_2 of VCU module.

3 Measure the resistance between the harness terminal of brake pedal positioning sensor and the VCU.

Measuring circuit: resistance between terminal 1 on plug EB1010 and terminal N5 on plug EB1000_2.

Measuring circuit: resistance between terminal 2 on plug EB1010 and terminal D2 on plug EB1000_1.

Measuring circuit: resistance between terminal 3 on plug EB1010 and terminal D3 on plug EB1000_1.

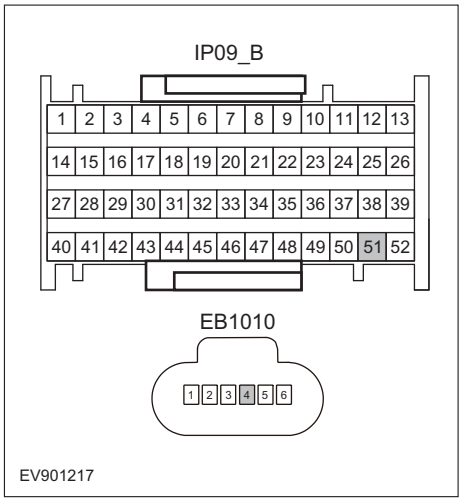
Standard value: ≈ 0Ω

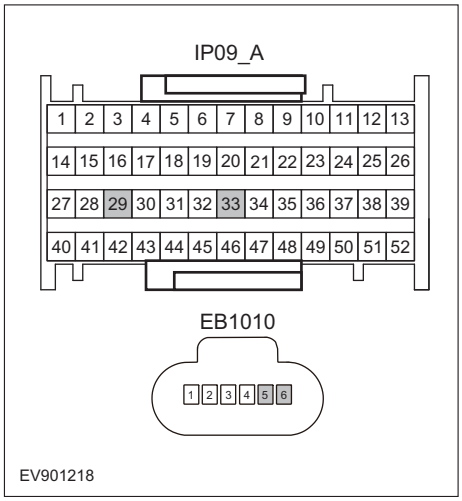
4 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 5	Check the circuit between brake pedal positioning sensor and BCM (open).





1 Turn the ignition switch to OFF.

2 Disconnect the harness plug IP09_A, IP09_B of BCM module.

3 Measure the resistance between the harness terminal of brake pedal positioning sensor and the BCM.

Measuring circuit: resistance between terminal 4 on plug EB1010 and terminal 51 on plug IP09_B.

Measuring circuit: resistance between terminal 5 on plug EB1010 and terminal 33 on plug IP09_A.

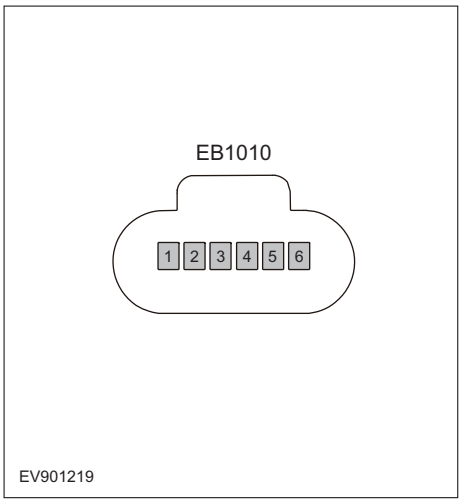
Measuring circuit: resistance between terminal 6 on plug EB1010 and terminal 29 on plug IP09_A.

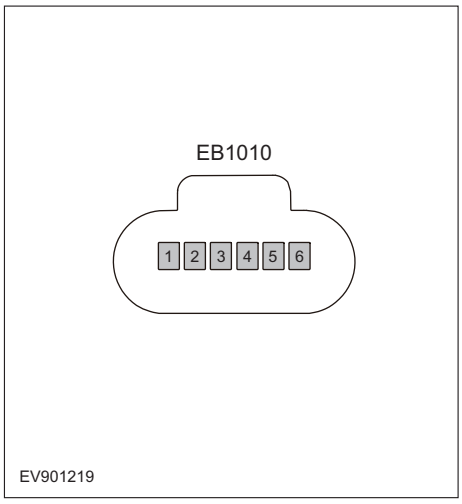
Standard value: ≈ 0Ω

4 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 6	Check the power supply circuit of brake pedal positioning sensor circuit (short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness plug of brake pedal positioning sensor and ground. Measuring circuit: resistance between terminal 1 on plug EB1010 and ground. Measuring circuit: resistance between terminal 2 on plug EB1010 and ground. Measuring circuit: resistance between terminal 3 on plug EB1010 and ground. Measuring circuit: resistance between terminal 4 on plug EB1010 and ground. Measuring circuit: resistance between terminal 5 on plug EB1010 and ground. Measuring circuit: resistance between terminal 6 on plug EB1010 and ground. **Standard value: ∞** 3 Measure the voltage between the harness plug of brake pedal positioning sensor and ground. Measuring circuit: voltage between terminal 1 on plug EB1010 and ground. Measuring circuit: voltage between terminal 2 on plug EB1010 and ground. Measuring circuit: voltage between terminal 3 on plug EB1010 and ground. Measuring circuit: voltage between terminal 4 on plug EB1010 and ground. Measuring circuit: voltage between terminal 5 on plug EB1010 and ground. Measuring circuit: voltage between terminal 6 on plug EB1010 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the brake pedal position sensor (components).
	1 Measure the resistance of the terminal of brake pedal position sensor directly. Measuring circuit: resistance between terminal 2 and terminal 3 on plug EB1010. **Standard value:** Measuring circuit: resistance between terminal 4 and terminal 6 on plug EB1010. **Standard value:** 2 Measure the resistance between the brake pedal position sensor terminal and the housing. Measuring circuit: resistance between terminal 1 on plug EB1010 and the housing. Measuring circuit: resistance between terminal 2 on plug EB1010 and the housing. Measuring circuit: resistance between terminal 3 on plug EB1010 and the housing. Measuring circuit: resistance between terminal 4 on plug EB1010 and the housing. Measuring circuit: resistance between terminal 5 on plug EB1010 and the housing. Measuring circuit: resistance between terminal 6 on plug EB1010 and the housing. **Standard value: ∞** 3 Check whether the result is normal or not?
	NO → Replace brake pedal position sensor.
YES ↓	
Step 8	Test whether VCU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the VCU Module

NO ↓
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.
Next Step ↓
Diagnosis end.